

Due: Monday, 17 November 2025, 11:59 PM

This is a group assignment, consisting of 2 or 3 students in each group.

Choose a suitable dataset from this repository:

<https://archive.ics.uci.edu/>

Make sure the dataset that you choose have more than 10 variables, and the response variable should be quantitative.

By using the Ridge and LASSO regressions, together with k-fold cross-validation, identify the best model for that dataset.

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DATASET : [Appliances Energy Prediction - UCI Machine Learning Repository](#)

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CHAPTER I

INTRODUCTION

1.1 DATA DESCRIPTION

The *Appliances Energy Prediction* dataset from the UCI Machine Learning Repository was selected for this assignment (Candanedo & Feldheim, 2017). The dataset contains measurements from a household, where energy consumption of appliances (in Wh) was recorded along with various environmental and weather-related variables. The dataset consists of 19,735 observations and 28 variables, including indoor temperature and humidity from multiple rooms, as well as external environmental conditions such as visibility, wind speed and pressure.

The response variable for this study is Appliances, representing the energy consumption of household appliances. Since this variable is quantitative, the dataset is suitable for applying Ridge and LASSO regression models. The presence of multiple predictors and possible multicollinearity makes this dataset appropriate for comparing regularization techniques combined with k-fold cross-validation to determine the best predictive model.

1.2 DATA VARIABLES

Variables in this dataset are described in the UCI repository. The target variable for this study is Appliances, which is the energy consumption of household appliances (in Wh). The variables used in this project are summarized in Table 1.

Table 1 Table of Data Variables

Variable	Role	Type	Units	Description
date	Feature	Date		Timestamp for each observation (YYYY-MM-DD HH:MM:SS).
Appliances	Target	Numeric (Continuous)	Wh	Energy use.
lights	Feature	Numeric (Continuous)	Wh	Energy use of light fixtures in the house.
T1	Feature	Numeric (Continuous)	°C	Temperature in the kitchen area.
RH_1	Feature	Numeric (Continuous)	%	Humidity in the kitchen area.
T2	Feature	Numeric (Continuous)	°C	Temperature in the living room area.
RH_2	Feature	Numeric (Continuous)	%	Humidity in the living room area.
T3	Feature	Numeric (Continuous)	°C	Temperature in the laundry room area.
RH_3	Feature	Numeric (Continuous)	%	Humidity in the laundry area.
T4	Feature	Numeric (Continuous)	°C	Temperature in the office room.
RH_4	Feature	Numeric (Continuous)	%	Humidity in the office room.
T5	Feature	Numeric (Continuous)	°C	Temperature in the bathroom.
RH_5	Feature	Numeric (Continuous)	%	Humidity in the bathroom.
T6	Feature	Numeric (Continuous)	°C	Temperature outside the building (north side).
RH_6	Feature	Numeric (Continuous)	%	Humidity outside the building (north side)

T7	Feature	Numeric (Continuous)	°C	Temperature in the ironing room.
RH_7	Feature	Numeric (Continuous)	%	Humidity in the ironing room.
T8	Feature	Numeric (Continuous)	°C	Temperature in teenager room 2.
RH_8	Feature	Numeric (Continuous)	%	Humidity in teenager room 2.
T9	Feature	Numeric (Continuous)	°C	Temperature in the parents room.
RH_9	Feature	Numeric (Continuous)	%	Humidity in the parents room.
T_out	Feature	Numeric (Continuous)	°C	Temperature outside (from Chievres weather station)
Press_mm_hg	Feature	Numeric (Continuous)	mmHg	Atmospheric pressure (from Chievres weather station)
RH_out	Feature	Numeric (Continuous)	%	Humidity outside (from Chievres weather station)
Windspeed	Feature	Numeric (Continuous)	m/s	Wind speed (from Chievres weather station)
Visibility	Feature	Numeric (Continuous)	km	Outside visibility (from Chievres weather station)
Tdewpoint	Feature	Numeric (Continuous)	°C	Dew point temperature (from Chievres weather station)
rv1	Feature	Numeric (Continuous)		Random variable 1, nondimensional
rv2	Feature	Numeric (Continuous)		Random variable 2, nondimensional

1.3 OBJECTIVE

The objective of this assignment is to develop a predictive model that estimates household appliance energy consumption using the Appliances Energy Prediction dataset. Two techniques which are Ridge regression and Lasso regression, are applied to identify the most suitable model for predicting the target variable, Appliances. To ensure that the model performs reliably and avoids overfitting, k-fold cross-validation is used in the evaluation process. The goal is to determine which method provides the best predictive performance while identifying the most relevant variables influencing energy usage.

CHAPTER II

METHODOLOGY

2.1 LIBRARIES

2.1.1 Data Manipulation

a) Numpy (numpy)

Numpy was used to manage multi-dimensional arrays and carry out numerical operations, offering mathematical calculation necessary for data processing (Oliphant, 2006).

b) Pandas (pandas)

Pandas simplified the manipulation of tabular data, allowing for data organizing, cleaning and exploration before modelling (McKinney, 2010).

2.1.2 Data Preprocessing and Modelling

a) StandardScaler from sklearn.preprocessing

StandardScaler was used to standardize numerical variables in order to ensure that all predictors were on a same scale throughout model training by eliminating the mean and scaling to unit variance (Pedregosa et al., 2011).

b) KFold Cross Validation from sklearn.model_selection

The dataset was divided into several folds using KFold cross validation, which increased the model's evaluation and reduced the risk of overfitting (Pedregosa et al., 2011).

c) RidgeCV & LassoCV from sklearn.linear_model

Regularized regression models were fitted using RidgeCV and LassoCV. By using cross-validation to automatically choose the ideal regularization parameter α , these functions improved prediction accuracy while lowering model complexity (Pedregosa et al., 2011).

d) Mean_squared_error & r2_score from sklearn.metrics

The R2 score, which represents the percentage of variance explained by the model, and mean squared error (MSE), which measures the size of prediction error, were used to evaluate the model's performance (Pedregosa et al., 2011).

2.1.3 Data Visualization

a) Matplotlib (matplotlib.pyplot)

To understand model behavior and assess performance trends, Matplotlib used to create visualizations including scatter plots and model coefficient plots (Hunter, 2007).

2.2 DATA PREPROCESSING

To make sure the dataset is clean, consistent and prepared for modelling, data preparation is a crucial step. The original dataset was examined for unnecessary features, duplication and missing values. To prevent bias and errors during model training, all rows with missing values were eliminated. This follows standard practice, as machine learning models generally perform poorly with incomplete data unless imputation techniques are applied (García et al., 2016).

Two variables, rv1 and rv2 were then eliminated since they are random variables that do not contribute meaningful information to the model's prediction. Eliminating irrelevant variables helps improve model interpretability and avoids unnecessary variance in the modelling process (Kuhn & Johnson, 2013).

2.3 DATA MODELLING

Ridge Regression and Lasso Regression were the two regularized linear regression models that applied. Both models aim to improve generalization and avoid overfitting by penalizing large coefficients.

2.3.1 Ridge Regression

Coefficient magnitudes are shrunk by Ridge Regression, but they are not forced to zero. This approach is effective when predictors are correlated, as the penalty distributes weights more evenly across features. The model was built using RidgeCV, which automatically searches for the best regularization parameter (α) by assessing several predictors and selecting the one with lowest error (Hoerl & Kennard, 1970).

2.3.2 Lasso Regression

Lasso Regression performs automatic feature selection by introducing an L1 penalty that can shrink some coefficients to zero. This is especially helpful when there may be repetitive or unimportant predictors in the dataset. LassoCV, which similarly finds the ideal α that minimizes validation error, was used to develop the model (Tibshirani, 1996).

2.4 EVALUATION METRICS

2.4.1 K-Fold Cross Validation

5-fold cross-validation was used to guarantee the model evaluation's stability and robustness. Five equal subsets were partitioned from the dataset. 4 folds were used for training and the remaining fold was used as a test set for each model. This process was repeated five times so that each fold was used once as a test set. By averaging the outcomes from every iteration, the

average Negative Mean Squared Error was computed to assess the model, increasing generalization, avoiding overfitting, and enhancing performance reliability.

2.4.2 Negative Mean Squared Error (neg-MSE)

As specified by scikit-learn's scoring convention, Negative Mean Squared Error (neg-MSE) was the performance metric used during cross-validation. Higher scores are indicative of superior performance because of the negative value, which is consistent with other measures like accuracy. The neg-MSE data were transformed back to MSE for simpler reading following cross-validation. Lower MSE values are indicated for greater model performance since MSE measures the average squared difference between predicted and actual values.

CHAPTER III

RESULTS AND DISCUSSION

3.1 MODEL PERFORMANCE

This chapter presents the performance of Ridge and Lasso regression models in predicting appliance energy consumption. Both models were evaluated using 5-fold cross-validation, and their performance was assessed based on mean squared error (MSE), R^2 values, and coefficient behaviour. The following subsections describe the performance of each model and provide visual diagnostics through plots. A comparison is then made to identify the best model, followed by an evaluation of the selected model's predictive structure.

3.1.1 Predicted vs Actual Appliance Energy Use

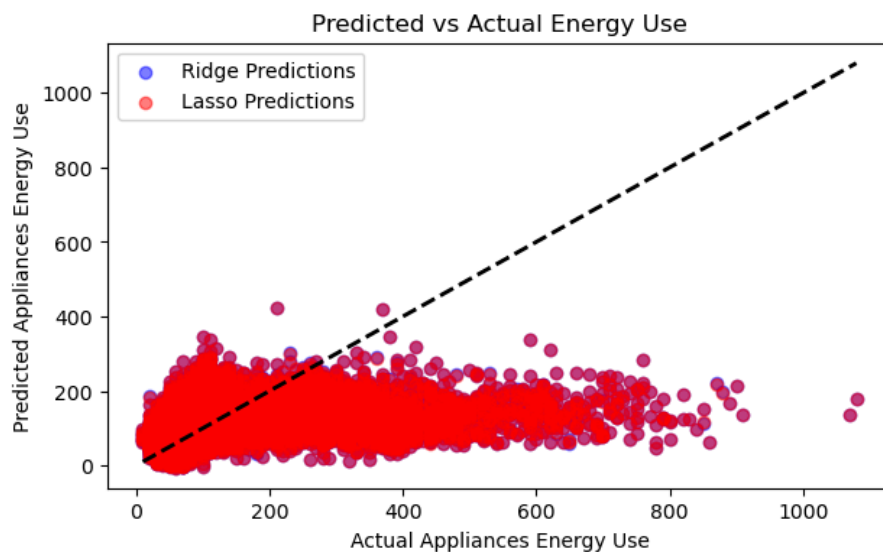


Figure 1 *Predicted vs Actual Appliance Energy Use (Ridge and Lasso Regression)*

Figure 1 illustrates the predicted versus actual appliance energy usage for both Ridge and Lasso regression models. Actual appliance energy consumption ranges from close to 0 up to more than 1000 Wh, while both models produce predictions primarily between 0 and 400 Wh. This results in a visible underestimation for higher consumption levels, where predicted values fall below the diagonal reference line. The dispersion of residuals widens as actual appliance usage increases, indicating that the models perform better for low to moderate consumption ranges and struggle to represent extreme peaks effectively. Overall, the scatter plot suggests that both models capture general trends but exhibit limited ability to predict high appliance energy usage accurately.

3.1.2 Ridge Regression

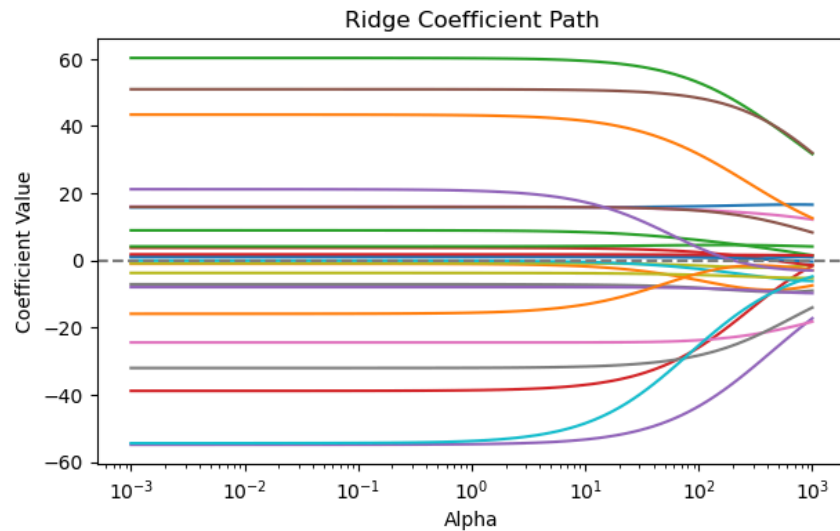


Figure 2 *Ridge Coefficient Path*

Ridge regression was fitted using a range of regularisation strengths, and the optimal α was determined using 5-fold cross-validation. Figure 2 shows the Ridge coefficient path as α increases from 0.001 to 1000. At very small α values, several coefficients reach large magnitudes, such as RH_1 at approximately +60 and RH_2 at approximately -57. As α increases, the coefficients gradually shrink while remaining non-zero, reflecting Ridge's continuous shrinkage behaviour.

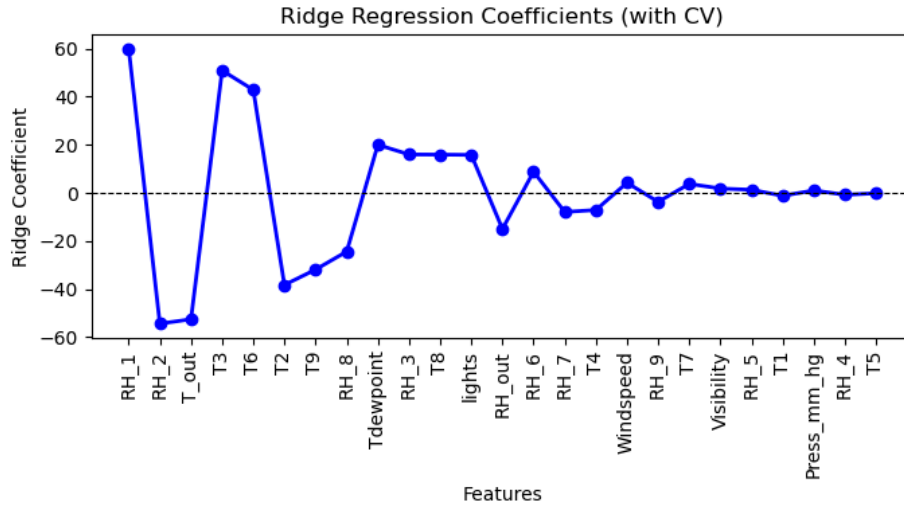


Figure 3 Ridge Regression Coefficients (with CV)

Figure 3 presents the final Ridge coefficients obtained using the optimal α value. The strongest predictors in the model include RH_1 (+60.037), T3 (+50.901), and T6 (+42.848). These variables exhibit the largest magnitudes and therefore exert the greatest influence on appliance energy consumption. Predictors with much smaller magnitudes, such as RH_5 (+1.286) and Press_mm_hg (+1.030), contribute minimally to the predictions.

3.1.3 Lasso Regression

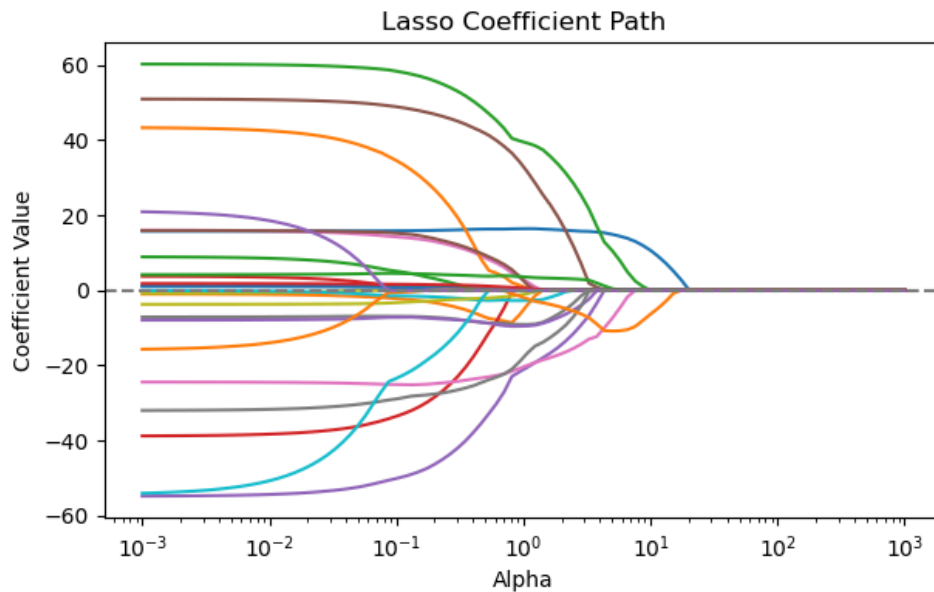


Figure 4 Lasso Coefficient Path

The Lasso regression model was also trained across a range of α values with 5-fold cross-validation. Figure 4 shows the coefficient path. When α is small, the coefficients display magnitudes similar to those observed in Ridge regression. As α increases, however, Lasso coefficients shrink more aggressively, and several approach zero at higher regularisation strengths. Since the optimal α for Lasso is 0.01417, the final model displays mostly moderate shrinkage rather than extensive sparsity.

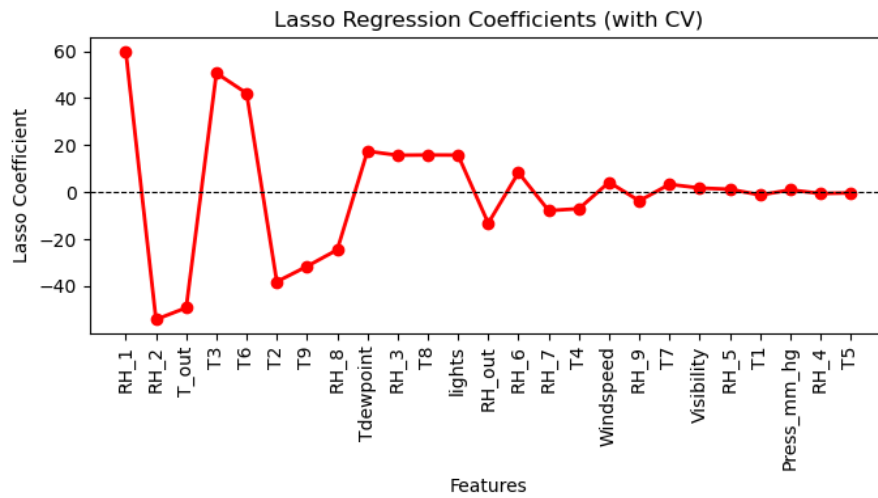


Figure 5 *Lasso Regression Coefficients (with CV)*

Figure 5 shows the final Lasso coefficients obtained using the optimal α value. Major predictors such as RH_1 (+60.029), RH_2 (-54.161), T3 (+50.688), and T6 (+42.121) remain strong contributors. Smaller coefficients, such as Press_mm_hg (+0.993), experience more shrinkage in Lasso compared to Ridge, demonstrating Lasso's stronger effect on less influential predictors.

3.2 COMPARISON BETWEEN MODELS

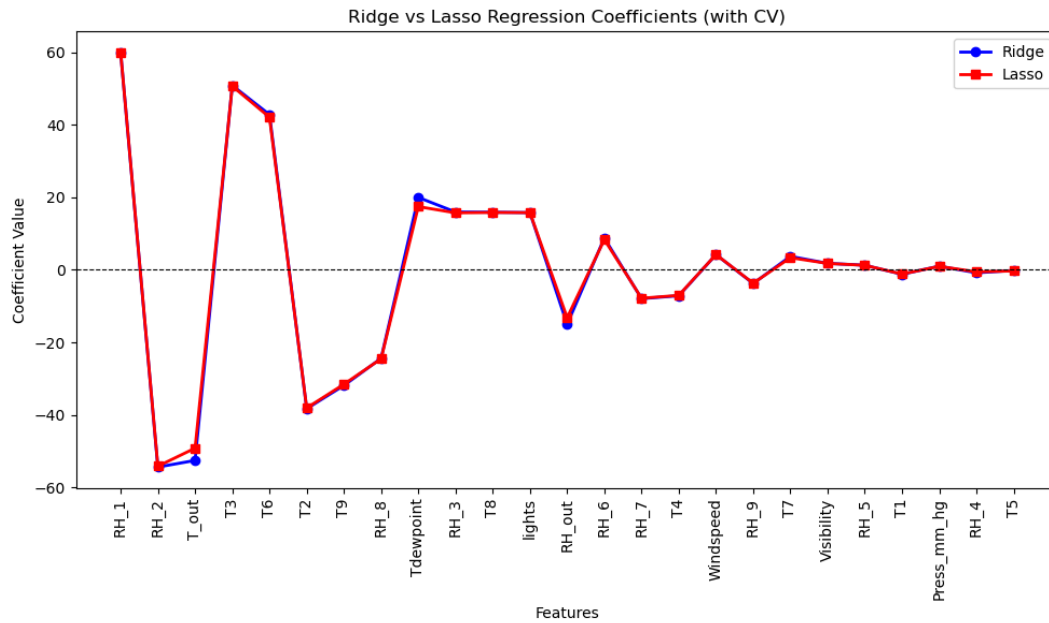


Figure 6 Ridge vs Lasso Regression Coefficients (with CV)

Figure 6 compares the Ridge and Lasso Regression coefficients. Both models produce nearly identical coefficient structures, especially among the most influential predictors. For example, RH_1 has coefficients of +60.037 (Ridge) and +60.029 (Lasso), while T3 has +50.901 (Ridge) and +50.688 (Lasso). Differences are more noticeable for smaller coefficients, where Lasso applies more shrinkage. The similarity between both models' coefficient patterns indicates that they are capturing the same underlying relationships in the dataset.

3.3 SELECTING BEST MODEL

Table 2 Comparison of Model Performance

Model	Number of Features	Best alpha	MSE	R^2
Ride Regression	25	2.84804	8777.9153	0.16487
Lasso Regression	25	0.01417	8778.1201	0.16485

In evaluating the best predictive model between Ridge and Lasso regression, key performance metrics such as the mean squared error (MSE) and the coefficient of determination (R^2) were used. Both models were trained using 5-fold cross-validation, and regularisation strengths (α values) were optimised automatically based on prediction error.

Ridge regression achieved the lowest MSE value of 8777.9153, indicating slightly better prediction accuracy compared to Lasso regression, which recorded an MSE of 8778.1201. Ridge also produced a marginally higher R^2 value of 0.16487, suggesting that it explains a slightly greater proportion of variation in appliance energy consumption. Although the performance difference is small, it consistently favours Ridge across both key evaluation metrics.

Lasso regression, with a selected α of 0.01417, produced performance metrics very close to those of Ridge. While Lasso is typically useful for enforcing sparsity and reducing the number of active predictors, the selected α value in this dataset did not lead to significant coefficient elimination. As a result, both Ridge and Lasso effectively used the same set of predictors, diminishing the usual sparsity advantage of the Lasso method.

Given its slightly superior MSE and R^2 scores and its stability when dealing with the highly correlated environmental variables present in the dataset, Ridge Regression is justified as the best model for this study. Its consistent performance across metrics and its ability to retain all predictors while controlling coefficient shrinkage make it the most suitable approach for modelling appliance energy consumption.

3.4 EVALUATING BEST MODEL

Table 3 **Coefficient Estimates for Ridge and Lasso Regression Models (with CV)**

X	Feature	Ridge Coefficient	Lasso Coefficient
X1	lights	15.80327	15.79680

X2	T1	-1.20831	-1.16042
X3	RH_1	60.0371	60.02941
X4	T2	-38.29143	-38.08343
X5	RH_2	-54.39309	-54.16072
X6	T3	50.90121	50.68781
X7	RH_3	15.97501	15.74055
X8	T4	-7.12573	-7.02866
X9	RH_4	-0.80284	-0.55139
X10	T5	-0.15699	-0.28250
X11	RH_5	1.28558	1.27491
X12	T6	42.84803	42.12076
X13	RH_6	8.79866	8.35467
X14	T7	3.73899	3.36643
X15	RH_7	-7.93553	-7.82254
X16	T8	15.91419	15.84846
X17	RH_8	-24.40638	-24.52118
X18	T9	-31.88611	-31.54922
X19	RH_9	-3.74626	-3.67798
X20	T_out	-52.56569	-49.18262
X21	Press_mm_hg	1.03030	0.993368
X22	RH_out	-14.99923	-13.18730
X23	Windspeed	4.20889	4.24263
X24	Visibility	1.81017	1.78739
X25	Tdewpoint	19.99399	17.46394

As Ridge regression was identified as the most suitable model, the coefficient estimates were examined to identify the variables that most significantly influence appliance energy consumption. The Ridge model retains all 25 predictors, and the magnitude and direction of the coefficients provide insight into the relationship between indoor environmental conditions and household energy use.

Several predictors emerged with notably large absolute coefficients, indicating a strong impact on appliance consumption. Among these, indoor humidity and temperature measurements were dominant. For instance, RH_1 (coefficient $\approx +60.037$) and T3 ($\approx +50.901$) were the strongest positive contributors, suggesting that increases in humidity and temperature in certain rooms correspond to higher appliance usage. Conversely, RH_2 displayed a large negative coefficient (≈ -54.393), reflecting an inverse relationship. Additional influential predictors included RH_8 and RH_4, both with significant positive magnitudes. These findings imply that indoor comfort conditions, particularly humidity gradients and temperature variations across different sensor locations play a central role in determining household energy demand.

Predictors with moderate coefficients, such as RH_6, RH_7, RH_out, and Tdewpoint, also contributed relevantly to the model. These variables capture fluctuations in indoor and outdoor humidity patterns, dew point levels, and transitional climate factors that may indirectly affect energy usage through behavioural or environmental adjustments within the home. In contrast, features such as RH_5, Press_mm_hg, T4, and Visibility exhibited smaller magnitudes, indicating limited influence relative to the major predictors. While still retained in the model, their contribution to overall predictive power is modest.

Overall, the Ridge model reveals that appliance energy consumption is strongly driven by thermal and humidity conditions within the household, rather than by external environmental factors alone. The diverse range of influential features such as spanning indoor temperatures, relative humidity readings and selected outdoor climate indicators highlights the multifaceted nature of household energy behaviour. While Lasso imposes slightly stronger shrinkage on the smaller coefficients, Ridge retains all predictors in the model. As a result, the Ridge model

provides a more complete and stable depiction of the indoor environmental factors associated with household energy usage.

Using the estimated Ridge coefficients, the predictive model for appliance energy consumption can be written as:

$$\begin{aligned}\hat{y} = & 15.80327x_1 - 1.20831x_2 + 60.0371x_3 - 38.29143x_4 - 54.39309x_5 \\ & + 50.90121x_6 + 15.97501x_7 - 7.12573x_8 - 0.80284x_9 - 0.15699x_{10} \\ & + 1.28558x_{11} + 42.84803x_{12} + 8.79866x_{13} + 3.73899x_{14} - 7.93553x_{15} \\ & + 15.91419x_{16} - 24.40638x_{17} - 31.88611x_{18} - 3.74626x_{19} - 52.56569x_{20} \\ & + 1.03030x_{21} - 14.99923x_{22} + 4.20889x_{23} + 1.81017x_{24} + 19.99399x_{25}\end{aligned}$$

where each X_i corresponds to a specific predictor variable from the dataset.

The equation summarises how each environmental variable contributes to predicting appliance energy usage. Positive coefficients indicate that increases in the respective variables are associated with higher appliance consumption, whereas negative coefficients reflect inverse relationships. This final model provides an interpretable and stable representation of the environmental conditions driving household energy demand.

CHAPTER IV

CONCLUSION

This study evaluated the performance of Ridge and Lasso regression models in predicting household appliance energy consumption using environmental and weather-related variables. Both models were assessed using 5-fold cross-validation, and the results showed that their predictive accuracy was relatively similar. However, Ridge regression achieved the lowest MSE (8777.9153) and the highest R^2 (0.16487), outperforming Lasso (MSE = 8778.1201, R^2 = 0.16485). The small performance difference, together with the high multicollinearity among indoor temperature and humidity features, demonstrates Ridge's advantage in stabilising coefficients without removing predictors. Lasso, despite its capability for variable selection, retained all predictors due to the small optimal α value (0.01417), resulting in coefficient patterns closely resembling Ridge.

Analysis of the estimated coefficients highlighted that indoor environmental conditions, particularly RH_1, RH_2, T3, and T6, exerted the strongest influence on appliance energy consumption, whereas variables such as Press_mm_hg, T4, and Visibility played relatively minor roles. These findings suggest that fluctuations in indoor humidity and temperature are key drivers of household energy usage. Overall, Ridge regression emerged as the most appropriate model for this dataset, offering slightly better predictive performance and greater coefficient stability. This reinforces the value of regularised regression method, supported by cross-validation as reliable tools for modelling complex, multivariate energy-consumption behaviour.

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